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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 523 — Past Questions & Materials

by Renance

Course: PEE 523

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 3

Levels: 500

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Paper 1 of 3 — PEE 523 524 526 PAST QUESTIONS 2024/2025

Level 500 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
DEPARTMENT OF PETROLEUM ENGINEERING
PRODUCTION ENHANCEMENT AND PRODUCTION OPTIMISATION, 2023/2024

COURSE CODE: PEE 523

Answer ALL Questions

Time: 3 Hours

- 1a) Explain the terms "Production Enhancement", Production Optimisation, and "Well Intervention". State the main elements of a production system.
- b) How is gas well deliverability assessed? Why is it important?
- c) The average pressure of a reservoir is given as 2 300 psi, with flowing wellbore pressure as 1 400 psi, and oil production rate of 95 bbl/d.
Calculate the oil rate when the flowing wellbore pressure is 700 psi.
- 2a) Pressure loss in a production system is highest in the tubing. Explain.
- b) Explain the terms: Flow regimes, Non-Newtonian fluids (Use petroleum engineering examples and diagrams to illustrate) and a sick well.
- b) Using pressure gradient curve, determine the flowing bottom hole pressure for vertical tubing multiphase production.
Given Data: Bottom hole pressure = 2,5000 psig, Tubing size = 2.5 in. (ID). Water specific gravity = 1.074, Average temp = 140 deg. F, GLR = 400 scf/STB, Tubing length = 8,000 ft and Oil rate = 500 bpd, Oil gravity = 35 deg. API.
- d) Mention the three (3) key segments in subsurface production that need to be optimized for effective well productivity?
- 3a) Raise a team for a gas well stimulation.
- b) A welltest shows that the flow efficiency of a well is 0.4. Kindly write a proposal to your production supervisor for any action that may be necessary.
- c) Write a short note on Well Stimulation for a shaly carbonate formation.
- 4a) What are the applications of horizontal well production?
- b) What are the factors responsible for increase in global energy demand?
- c) Explain the term "energy transition" and "gas is a viable transition fuel".
- 5a) Why are production chokes installed at the wellhead? When can a choke be found at the tubing shoe?
- b) The wellhead pressure of a well is given as 182 psig and the flowrate, 200 bbl/d with a GOR of 700 scf/bbl. Calculate the bean size that will flow this well.
- C) What is the purpose of artificial lift and benefits of gas lift application? How is gas lift different from gas flooding? Mention the three main types of ALM.

PRODUCTION ENHANCEMENT AND PRODUCTION OPTIMIZATION

2023/2024

1a) Production enhancement refers to techniques and strategies used to increase the amount of hydrocarbons (oil or gas) extracted from a reservoir.

Production Optimization considers cost - It involves continuously improving and managing oil/gas well performance to maximize hydrocarbon recovery and reduce operational costs over the life of the well or field.

Well Intervention: It refers to any operation carried out on a well during or after its productive life that alter the state of the well or well geometry, provides diagnostics, or restore production.

Wellbore operation is part of well intervention.

Main elements of a Production System includes:

- ① The Reservoir ② The well ③ Production facilities.

4b) The Fetkovich equation is used to assess gas well deliverability and performance.

It is important to assess well deliverability to:

- ① Determine the amount of gas a well can produce on demand.
- ② The deliverability test is a requirement for regulatory body.
- ③ Used to forecast long term behaviour of wells & Reservoir.

Average reservoir pressure = 2300 psi, $P_{wf} = 1400$ psi

$q_o = 95$ bbl/d

Find $q_o = ?$ at $P_{wf} = 700$ psi

$$\frac{q_o}{q_{max}} = 1 - 0.2 \left(\frac{P_{wf}}{P_e} \right) - 0.8 \left(\frac{P_{wf}}{P_e} \right)^2$$

$$\frac{q_o}{q_{max}} = 1 - 0.2 \left(\frac{1400}{2300} \right) - 0.8 \left(\frac{1400}{2300} \right)^2 = 0.581$$

$$q_{\max} = \frac{q_0}{0.4} = \frac{95}{0.58} = 163.51 \text{ stb/d}$$

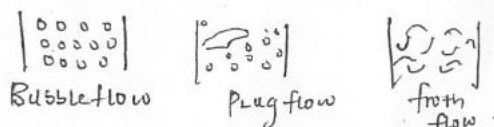
$$\frac{q_0}{q_{\max}} = 1 - 0.2 \left(\frac{q_0}{2300} \right) - 0.8 \left(\frac{q_0}{2300} \right)^2$$

$$\frac{q_0}{163.51} = 0.8650$$

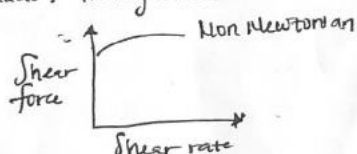
$$q_0 = 0.8650 \times 163.51 = 141.432 \text{ stb/day}$$

2a) In a production system, pressure loss is highest in the tubing primarily due to long length and relatively small diameter of the tubing combined with frictional losses as the fluid flows from the reservoir to the surface.

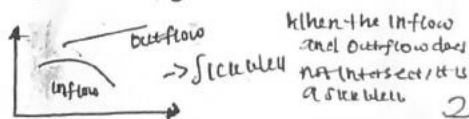
2b) flow regime is the relationship between forces in fluids (e.g.) Reynolds number. Define the relationship between forces.



Non newtonian fluids are fluid whose viscosity changes with the applied shear rate (unlike Newtonian fluids whose viscosity remains constant) e.g. Drilling muds, Heavy crude oil.



A Slick well refers to a producing well that is underperforming relative to its potential due to issues like formation damage, lost or gas conning. A Slick well has a Productivity Index (PI) less than 10% of original value.

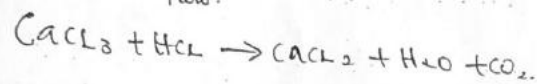


4) Acid fracturing - To go below the fractures.

3) For Shale Carbonate formation, the use of Hydrochloric acid (HCl) is recommended;

a) The acid reacts with the carbonate.

b) The process is exothermic (heat liberated), the oil viscosity is reduced and therefore enhance flow.



4a) Applications of horizontal well production includes;

① Enhanced Oil Recovery: Increases contact with the reservoir, boosting production.

② Tight Reservoirs: Useful in shale or low permeability formations.

③ Water/gas coning control: Reduces unwanted water/gas production.

4b) Factors Responsible for Increase in Global energy demand;

a) Population growth.

b) Technological advancement

c) Economic development

d) Industrialization.

4c) Energy transition is described as the global shift from fossil fuels (coal/oil) to cleaner energy sources (renewable/nuclear) to reduce carbon emissions and combat climate change.

Gas is a viable transition fuel because it emits less CO₂ than coal and oil. It's reliable, scalable and supports intermittent renewables (like solar/wind) during the transition.

5a) The production choke is installed at the wellhead to allow for flow restriction. The choke is designed for critical flow i.e. when the inlet pressure is about twice the outlet pressure.

When the choke is found at the tubing shoe, then there is possibility of Gas Hydrate.

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5b) $P_{\text{required}} = \text{THP} = 182 \text{ psi}$ $C = 1$ (Assume to be 1)

$q = 200 \text{ bbl/d}$

$R = 40 \text{ R} = 700 \text{ scf/bbl}$

$\text{THP} = \frac{C R^{0.5} q}{S^2}$, $S^2 = \frac{C R^{0.5} q}{\text{THP}}$

$S^2 = ?$

$S = \sqrt{\frac{C R^{0.5} q}{\text{THP}}}$

$S = \sqrt{\frac{1 \times (700)^{0.5} \times 200}{182}}$

$S = \sqrt{\frac{5291.5}{182}} = 5.39$

Convert to bore size (in 1/64th of an inch)

$S = 5.4 \times \frac{1}{64} = \frac{5}{64}$

The bore size is $5 - 6/64$.

5c) The Purpose of artificial lift is to increase reservoir pressure. It is used when reservoir pressure is low. It is an assisted flow method.

Benefits of Gas Lift

- ① The gas is injected at a pressure of 200 psi in excess of 250.
- ② The gas mixes with the oil, aerate it, so it lighten the fluid column.
- ③ At great depth, the gas expands by about 360 times the original volume, so the oil is produced.

The difference between gas flooding and gas lifting is that

* Gas flooding is for EOR (Enhanced Oil Recovery), it is injection of gas in to the reservoir, while gas lifting is gas injected in to the gas lift valve through the gas lift manifold.

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Types of Artificial Lift Method (ALM)

- ① Gas Lift.
- ② Beam or sucker rod Pumping System.
- ③ Electrical Submersible Pump (ESP).

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY

Department of Petroleum Engineering
2023/2024 Second Semester Examination

PEE 524: Integrated Field Development and Planning

Answer Questions 1 and any other three. Time Allowed: 2 Hrs. 30mins

- 1(a)i. Define 'field development planning (FDP)' and its importance.
ii. List the five stages of the oil field development process.
 - (b) i. Explain the term 'marginal field.' ()
ii. Discuss how SWOT analysis can influence field development decisions.
 - (c) i. You are tasked with estimating the reserves of a newly discovered oil field. Given the data from initial exploratory wells, you decide to apply deterministic modeling to estimate the recoverable oil reserves. The data you have includes the area of the reservoir (A) 2km², the average net pay thickness (h) 30m, porosity 20%, water saturation 25%, and the formation volume factor (Bo) 1.2rb/STB. NB: Convert all the given values into consistent units.
ii. Compare and contrast deterministic and stochastic reserves estimation, providing examples of when each might be preferred.
-
- 2(a)i. Mention any three information that must be defined and economically evaluated in a typical petroleum field development project.
ii. Detail the process and significance of choosing production system options for offshore field development.
 - (b)i. Today, why does a typical field development planning involves greater detailed studies? Mention at least 4 points.
ii. How do we achieve acreages?
 - (c). Describe the role of project management in the execution phase of the FDP.
-
- 3(a)i. Analyze the importance of new technologies in managing marginal fields.
 - (b)i. Explain the function of the design basis section of development planning and what are the parameters employed?
ii. State some (at least 7) important components of the design basis section.
 - (c). Explain the engineering section of development plan and mention the key elements.
-
- 4(a)i. Evaluate the potential impacts of not conducting an After-Action Review (AAR) after FDP execution.
ii. What is the purpose of an After-Action Review (AAR) in field development?
 - (b)i. Describe two criteria considered in field development planning (FDP). *Technical*
ii. What methods would you suggest to a team to improve the performance of a developed field? *Benchmarking, Regular Audits, Identify Improvement Gaps, Develop Improvement Project, Establish Metrics & Monitor*
 - (c) What are the planning horizons generally in FDP?
-
- 5(a)i. Having been part of an FDP team, how will you best implement subsurface realization?
ii. Why is it important to consider both subsurface and surface aspects in FDP?
 - (b)i. Explain the significance of 'farm out/farm in arrangement' in marginal field development.
ii. What are the major differences between oil and gas field development?
 - (c). Explain the significance of managing a marginal field.

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PEE 524 - INTEGRATED FIELD DEVELOPMENT AND PLANNING.

1a) Field development planning is a multidisciplinary and integrated approach of performing and analysing feasibility studies on the exploitation design basis, Engineering, operations and maintenance as well as economics aspect of developing an oil field.

1a) The five stages of oil field development process:

- ① Field discovery
- ② Field Evaluation
- ③ Field development
- ④ Field production
- ⑤ Field abandonment.

1b) A marginal field can be defined as (appraised or not) in the range of 5 - 20 million barrels reserve that has remained undeveloped for a fairly long period of time (usually 5 to 10 years) and fall on the border line between economic and uneconomic reserves hence they are economically sensitive.

1bii) SWOT analysis influences field development decision. SWOT analysis influences FDP because it is used to evaluate the strength, weakness, opportunities and threats involved in a project.

In FDP, SWOT analysis helps in identification and is essential in subsequent steps in the process of planning or development of objectives of the discovered field.

S-S

4a) OOIP = ?

$$\text{Reserve Volume} = (A \times h) \times \phi (1 - S_o)$$

$$A = 2 \text{ km}^2 \quad \phi = 20\% = 0.2 \quad B_o = 1.28 \text{ bbl/stb}$$

$$h = 30 \text{ m} \quad S_w = 25\% = 0.25$$

$$A = 2 \text{ km}^2$$

$$1 \text{ km}^2 = 247.105 \text{ Acres}$$

$$2 \text{ km}^2 = x$$

$$A = 2 \times 247.105 = 494.21 \text{ Acres}$$

$$h = 30 \text{ m}$$

$$1 \text{ m} = 3.28 \text{ ft}$$

$$30 \text{ m} = x$$

$$h = 30 \times 3.28 = 98.4 \text{ ft}$$

$$\text{OOIP} = \frac{7758 \times A \times h \times \phi (1 - S_o)}{B_o} = \frac{7758 \times 494.21 \times 98.4 \times 0.2 \times (1 - 0.25)}{1.2}$$

$$= \frac{56,661,460.99}{1.2}$$

$$= 47,167,884.16 \text{ STB}$$

$$\text{OOIP} = 47.17 \text{ mm STB}$$

4c) Deterministic Reserve Estimation involves calculating reserves using single best estimate values for input parameters (e.g. area, thickness, Porosity & saturation).

Stochastic estimation involves assigning probability distribution to input parameters and performing simulations (e.g. Monte Carlo simulation) to regenerate a range of possible outcomes.

2a) In a typical petroleum field development project, 3 information to be defined and economically evaluated includes,

(1) The number of wells to be drilled to reach production objectives

(2) The recovery techniques to be used to extract the fluid, within the reservoir.

(3) The separation system needed for gas and fluid.

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2bii) He can achieve acreage through,

a) Farm out / farm in arrangement: In this type of arrangement, the concessionaire is not interested in drilling out at a time but agrees to assign the lease of the field or portion of it to another operator or farminee who is ready to develop the acreage.

b) Interest Acquisition: In this case, the farminee carries out an agreed work programme in return for a percentage interest from the field. He takes responsibilities for all aspects of development including drilling, operations and financial.

c) Overriding Royalty: Here, the farmowner party assigns his interest to the farminee who then earn an overriding interest.

d) Tariff Payment: In this case, the farminee receives a fixed or adjustable tariff in dollar per barrel. The fee could also come as a percentage of Net Production.

2c) The role of Project management in the execution phase of FDP includes,

i) The appointment of Project leader to co-ordinate / schedule and forward Plan Preparation.

ii) A clear designation and allocation of responsibilities among team members helps to ensure that the plan is prepared in an efficient and timely manner.

iii) Ensuring approval and Review Limiting and Authority by Management.

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3a) The Importance of New technologies in managing marginal fields includes:

- i) Partnership and allowana concept,
- ii) Common Sharing of Production facilities.
- iii) Use of Total Quality Management [TQM]

3b) The design basis Section Summarizes the major parameters of development Planning and includes a vivid description of the aims and objectives of the field development planning.

* Major parameters are employed in the Engineering Section to arrive at detailed facilities and equipment requirements and specifications.

3bii) Important components of the design basis include:

- i) Reservoir and fluid characteristics
- ii) Production and fluid characteristics
- iii) Wells
- iv) Engineering, Operating and Maintenance Philosophy
- v) Installation
- vi) Transportation
- vii) Infrastructure.

3c) The Engineering Section Contains the detailed Engineering used to design the Production facility and transportation systems. It gives the details of the method used, major assumptions and options (or alternatives) investigated.

Key elements of the Engineering Section include:

- i) Field Installation and Drilling Facilities

- (2) Production facilities
- (3) Hydrocarbon Transport - System
- (4) Project organization.

4a) In a situation in which after ~~on~~ field development planning execution an After Action Review (AAR) is not conducted, it could lead to

- i) Potential issues not being resolved
- ii) Prevent Adequate Recommendation from being made.
- iii) Delay in executive decision board and stakeholder to proceed with project as scheduled.

4a) The Purpose of After Action Review in field development is for possible amendment.

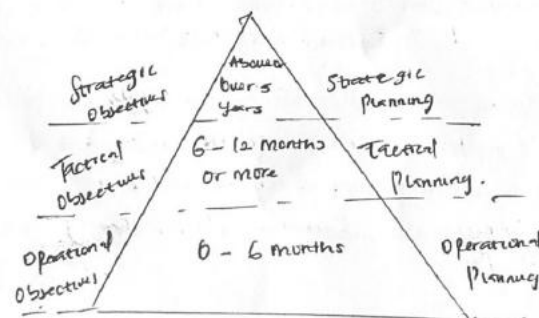
4b) The criteria considered in FDP include;

- ① Technical Consideration: It includes field location (Land or sea), field size, multiple reservoirs, ultimate recovery, reservoir attributes, crude quality (STOIP) and separator capacity.
- ② Economic Consideration: These include: Oil price, Economic Policies, field expenses, Capital cost, market control and constraints, taxation, project viability, demand for oil, royalties.
- ③ Commercial consideration
- ④ Operational consideration
- ⑤ Political Consideration.

- 4b) Methods of Improving field development performance;
- i) Benchmarking learn best practices (Compare your performance with those of outsiders)
 - ii) Identify performance gaps (i.e. gaps between expectations and actual)
 - iii) Develop Improvement Plans
 - iv) Establish Control and Monitoring Systems.

4c) Planning Horizons In fdp

- i) Long Range Planning (Strategic Planning): It is the system of an organization that elaborates an effort to improve overall effectiveness. It focuses on long-term horizon of more than a year, roughly about 5 years and above.
- ii) Medium Range Planning (Tactical Planning): It is usually a plan for 1 year, though some companies may consider 1 to 3 years.
- iii) Short-Range Planning: It is a plan for just 3 to 6 months. (Operational Planning).



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5a) To implement the best subsurface realization the development plan will contain:

- 1) History and status
 - a) discovery and appraisal
 - b) Licenses and operators involved
 - c) offset and operators and local infrastructure
 - d) offset and tool-like field comparison
- 2) Geology and geophysics
 - a) seismic data
 - b) Structural interpretation
 - c) Maps and diagram
- 3) Petrophysical and Reservoir (fluid properties):
 - a) Petrophysical interpretation
 - b) Routine core analysis at atmospheric and overburden pressure and temperature.
- 4) Reservoir (hydrocarbon in place)
- 5) Reservoir management.

5aii) Considering both the surface and subsurface aspect info P ensures the following:

- 1) Surface development concept reviewed entails whether field production should stand alone or be combined with that of another field.
- 2) Subsurface development concept reviewed entails whether the wells should be vertical, deviated, inclined or horizontal.

5bi) farm in / farmout arrangement: In this type of arrangement, the concessionaire is not interested in drilling out at a time but agrees to assign the lease of the field or a portion of it to another operator or farmine who is ready to develop the acreage.

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5b) Major difference between oil and gas field development includes:

i) Gas field development requires a lot of investment on infrastructure, the biggest being and equipment being pipelines and compressors.

ii) Gas field development involves Liquefied Natural Gas (LNG) or Gas to Liquid (GTL) plants

iii) Economic expense on gas field investment development can only be recovered in a long run compared to the short term recovery of expense on investment for oil field development.

5c) The significance of managing a marginal field is the need for higher return on marginal investment.

Also it should be noted that the development of some field that are uneconomical in some parts of the world (such as West Africa) due to high development cost may be profitable in some parts of the world (such as Gulf of Mexico) where they have low development cost.

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
SECOND SEMESTER EXAMINATION, 2023/2024 SESSION

PEE 526: RESERVOIR SIMULATION

TIME ALLOWED: 3HRS UNIT: 3

INSTRUCTIONS: ANSWER ALL QUESTIONS IN SECTION A & TWO IN SECTION B

SECTION A

18 MARKS

QUESTION 1

Given a single-phase fluid flow in a 3D horizontal reservoir discretized into 4x3x3 blocks. A well, producing at 150 stb/d is located in block (2,2,2). All grid blocks have the following properties:

$$k_x = 324 \text{ mD}, k_y = 250 \text{ mD}, k_z = 73 \text{ mD}, \Delta x = 380 \text{ ft}, \Delta y = 415 \text{ ft}, \Delta z = 40.5 \text{ ft}, \beta = 1.125 \text{ rb/stb}, \mu = 2.5 \text{ cp}, \rho = 55 \text{ lbm/ft}^3$$

- Draw the reservoir representation of the above discretized reservoir.
- Order the blocks in this reservoir using the engineering notation.
- Identify the interior and boundary blocks in this reservoir.
- Write the flow equation for block (2,2,2) without substitution of values.
- Repeat (iv) by substituting "r" available data into the equation.

QUESTION 2

17 MARKS

For a single-phase linear flow in 3D, using the conservation equation, show that the left-hand side is equal to the right-hand side of the equation below.

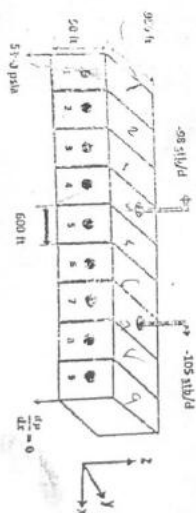
$$\begin{aligned} \Delta x \frac{\partial}{\partial x} \left(\beta_c \frac{k_{pAx}}{\mu} \left(\frac{\partial p}{\partial x} - \gamma_l \frac{\partial z}{\partial x} \right) \right) &+ \Delta y \frac{\partial}{\partial y} \left(\beta_c \frac{k_{pAy}}{\mu} \left(\frac{\partial p}{\partial y} - \gamma_l \frac{\partial z}{\partial y} \right) \right) \\ &+ \Delta z \frac{\partial}{\partial z} \left(\beta_c \frac{k_{pAz}}{\mu} \left(\frac{\partial p}{\partial z} - \gamma_l \frac{\partial z}{\partial z} \right) \right) \\ &= \frac{V_b}{\alpha_c} \rho \beta C_v \frac{\partial p}{\partial t} + q_{\text{net}} V_b \end{aligned}$$

15 MARKS

QUESTION 3

Consider the block-centred grid system in the figure below, with an initial reservoir pressure of 5300 psia, and a timestep of 12 days. Determine the pressure distribution during the first year of production, using the explicit formulation,

$$\begin{aligned} k_x &= 35 \text{ mD}, k_y = 28 \text{ mD}, k_z = 18 \text{ mD}, \Delta x = 380 \text{ ft}, \beta \\ &= 1.15 \text{ rb/stb}, \mu = 4.8 \text{ cp}, \phi = 0.20, C_v \\ &= 3.6 \times 10^{-6} \text{ psi}^{-1} \end{aligned}$$



QUESTION 4

SECTION B

10 MARKS

Discretize the 2D single phase linear reservoir system below using explicit formulation.

$$\Delta x \frac{\partial}{\partial x} \left\{ \beta_c \frac{k A_x}{\mu \beta} \left(\frac{\partial p}{\partial x} - \gamma' \frac{\partial z}{\partial x} \right) \right\} + q_{isc} = \frac{V_b \phi C_c \partial p}{\alpha_c \beta \partial t}$$

Where

$$T_{x, u/2} = \beta_c \frac{k A_x}{\mu \beta \Delta x} \quad \text{and} \quad \gamma'_{u/2} = \beta_c \frac{k A_y}{\mu \beta \Delta y}$$

QUESTION 5

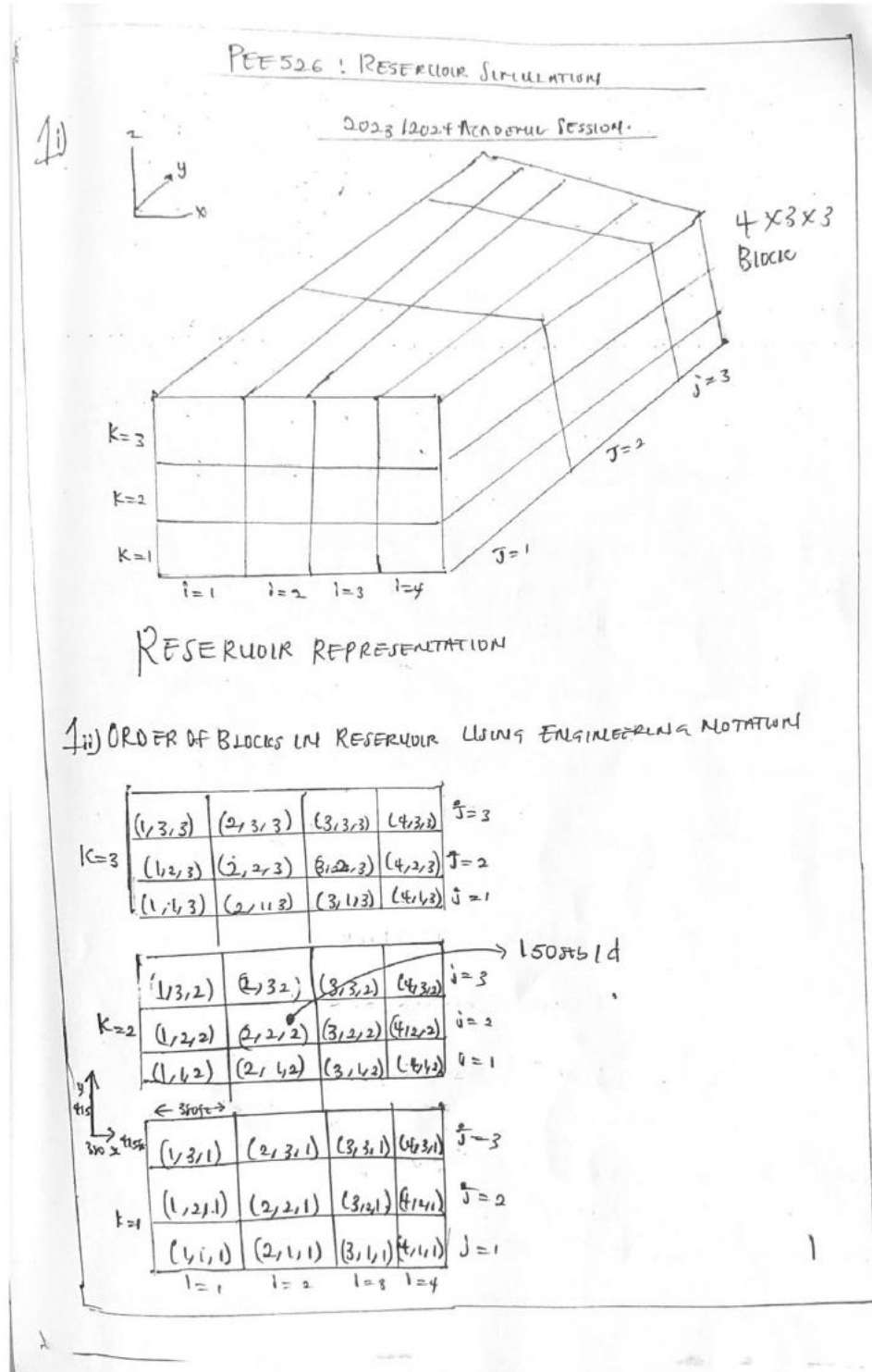
10 MARKS

- What are three different ways a reservoir can be modeled traditional
- State three data often used by a reservoir simulator to establish initial reservoir conditions
- State two source to obtain faults and boundaries data
- State four ways a reservoir can be modeled mathematically

QUESTION 6

10 MARKS

- State the two major classification or ways a reservoir can be modeled
- State two ways a reservoir can be modeled experimentally
- The transmissibility between two blocks is the measure of how easily fluids flow between them. Write the mathematical expression for two phase flow between grid block i and $i+1$, for water.
- State five factors that are considered and depended on for the diffusivity equation to be solved



4iii) As seen in question (2ii), interior blocks include reservoir blocks that are located on the second and third columns in the second row in the second layer i.e. blocks (2,2,2) and (3,2,2). All other reservoir blocks are boundary blocks.

4iv) To get the flow equation, considering Transmissibility

3D horizontal

$$\begin{aligned}
 & T_{x_{i,j,k-1/2}}^m (P_{i,j,k-1} - P_{i,j,k}) - y_{i,j,k-1/2}^m (Z_{i,j,k-1} - Z_{i,j,k}) \\
 & + T_{x_{i,j-1/2,k}}^m (P_{i,j-1,k} - P_{i,j,k}) - y_{i,j-1/2,k}^m (Z_{i,j-1,k} - Z_{i,j,k}) \\
 & + T_{x_{i-1/2,j,k}}^m (P_{i-1,j,k} - P_{i,j,k}) - y_{i-1/2,j,k}^m (Z_{i-1,j,k} - Z_{i,j,k}) \\
 & + T_{x_{i,j,k+1/2}}^m (P_{i,j,k+1} - P_{i,j,k}) - y_{i,j,k+1/2}^m (Z_{i,j,k+1} - Z_{i,j,k}) \\
 & + T_{x_{i,j+1/2,k}}^m (P_{i,j+1,k} - P_{i,j,k}) - y_{i,j+1/2,k}^m (Z_{i,j+1,k} - Z_{i,j,k}) \\
 & + T_{x_{i+1/2,j,k}}^m (P_{i+1,j,k} - P_{i,j,k}) - y_{i+1/2,j,k}^m (Z_{i+1,j,k} - Z_{i,j,k}) \\
 & = \frac{V_{i,j,k}}{\Delta t} \left(\left(\frac{\phi}{\beta} \right)_{i,j,k}^{n+1} - \left(\frac{\phi}{\beta} \right)_{i,j,k}^n \right)
 \end{aligned}$$

The flow Equation for block 2, 2, 2

$$\begin{aligned}
 & T_{z_{2,2,2-1/2}}^m [P_{2,2,2-1}^m - P_{2,2,2}^m] - y_{2,2,2-1/2}^m (Z_{2,2,2-1} - Z_{2,2,2}) \\
 & + T_{y_{2,2-1/2,2}}^m [P_{2,1,2}^m - P_{2,2,2}^m] - y_{2,2-1/2,2}^m (Z_{2,1,2} - Z_{2,2,2}) \\
 & + T_{x_{2-1/2,2,2}}^m [P_{1,2,2}^m - P_{2,2,2}^m] - y_{2-1/2,2,2}^m (Z_{1,2,2} - Z_{2,2,2})
 \end{aligned}$$

$$\begin{aligned}
 & + T_{x_{2+1/2,2/2}} [P_{3/2,2}^m - P_{2/2,2}^m] - Y_{2+1/2,2/2}^m (Z_{2/2,2} - Z_{2/2,2}) \\
 & + T_{y_{2/2+1/2,2}} [P_{2/3,2}^m - P_{2/2,2}^m] - Y_{2/2+1/2,2}^m (Z_{2/3,2} - Z_{2/2,2}) \\
 & + T_{z_{2/2,2+1/2}} (P_{2/2,3}^m - P_{2/2,2}^m) - Y_{2/2,2+1/2}^m (Z_{2/2,3} - Z_{2/2,2}) \\
 & + q_{sc_{2/2,2}}^m = \frac{Y_{2/2,2}}{\alpha_c \Delta t} \left[\left(\frac{\phi}{\beta} \right)_{2/2,2}^{n+1} - \left(\frac{\phi}{\beta} \right)_{2/2,2}^n \right] \quad \text{--- ①}
 \end{aligned}$$

for block (2,2,2)

$$2) \quad Z_{2/1,2} = Z_{1,2,2} = Z_{2/2,2} = Z_{3/2,2} = Z_{2/3,2}$$

$$Z_{2/2,1} - Z_{2/2,2} = 40.5 \text{ ft}$$

$$q_{sc_{2/2,2}}^m = 150 \text{ ft}^2/\text{s}$$

Because $\Delta x_{2-1/2,2/2} = 350 \text{ ft}$ $\Delta y_{2/2-1/2,2} = 415 \text{ ft}$

$\Delta z_{2/2,2-1/2} = 40.5 \text{ ft}$ and μ/ρ & β are constant

Then $Y_{2/2,2-1/2}^m = Y_{2/2,2+1/2}^m = Y_c \rho g = 0.21584 \times 10^{-3} \times 55 \times 32.174 = 0.3819 \text{ psi/ft}$

Note: $\mu = 2.5 \text{ cp}$, $\beta = 1.125$, $\Delta x = 40.5$, $\Delta y = 415$, $\Delta z = 350$

$K_x = 324 \text{ md}$, $K_y = 250 \text{ md}$, $K_z = 73 \text{ md}$

$A_x = \Delta y \Delta z = 415 \times 40.5 = 16807.5 \text{ ft}^2$

$A_y = \Delta x \Delta z = 350 \times 40.5 = 15390 \text{ ft}^2$

$A_z = \Delta x \Delta y = 350 \times 415 = 157170 \text{ ft}^2$

$\therefore A_{oc} = 16807.5 \text{ ft}^2$, $A_y = 15390 \text{ ft}^2$, $A_z = 157170 \text{ ft}^2$

3

$$T_{x_{2,1/2,2}}^m = \beta_c \frac{k_x A_x}{H \beta \Delta x} = \frac{0.001127 \times 324 (16807.5)}{2.5 (1.125) (50)} = 5.7424 \text{ stb/d-psi}$$

$$T_{y_{2,1/2,2}}^m = \beta_c \frac{k_y A_y}{H \beta \Delta y} = \frac{0.001127 \times 250 \times 15390}{2.5 \times 1.125 \times 45} = 3.7150 \text{ stb/d-psi}$$

$$T_{z_{2,1/2,2}}^m = \beta_c \frac{k_z A_z}{H \beta \Delta z} = \frac{0.001127 \times 73 \times 157700}{2.5 \times 1.125 \times 40.5} = 113.9019 \text{ stb/d-psi}$$

$$V_b = 350 \times 45 \times 40.5 = 613951.675 \text{ ft}^3$$

Substitute into equation 1

$$\begin{aligned} & (113.9019) (P_{2,2,1}^m - P_{2,2,2}^m) + 3.7150 (P_{2,1/2,1} - P_{2,1/2,2}) \\ & + 5.7424 (P_{1,2,2}^m - P_{2,1/2,2}^m) + 5.7424 (P_{3,2,2}^m - P_{2,1/2,1}^m) \\ & + 3.7150 (P_{2,1/2,2}^m - P_{2,2,2}^m) + 113.9019 (P_{2,2,1/3}^m - P_{2,2,1/2}^m) \\ & - 150 = \frac{V_b \Delta t}{\alpha_c} \left(\left(\frac{\psi}{\beta} \right)_{2,2,2}^{n+1} - \left(\frac{\psi}{\beta} \right)_{2,2,2}^n \right) \end{aligned}$$

② The general equation for 3D Single phase flow:

$$\begin{aligned} & \Delta x \frac{\partial}{\partial x} \left[\beta_c \frac{k P A_x}{H} \left(\frac{\partial P}{\partial x} - \gamma_1 \frac{\partial z}{\partial x} \right) \right] + \Delta y \frac{\partial}{\partial y} \left[\beta_c \frac{k P A_y}{H} \left(\frac{\partial P}{\partial y} - \gamma_1 \frac{\partial z}{\partial y} \right) \right] + \Delta z \frac{\partial}{\partial z} \left[\beta_c \frac{k P A_z}{H} \left(\frac{\partial P}{\partial z} - \gamma_1 \frac{\partial z}{\partial z} \right) \right] \\ & = \frac{V_b}{\alpha_c} \rho \phi C_t \frac{\partial P}{\partial t} + q_{\text{net}} V_b \end{aligned}$$

3D single phase.

$$[p_v]_x + [p_v]_y + [p_v]_z + q_{in} V_b - [p_v]_{x+\Delta x} + [p_v]_{y+\Delta y} + [p_v]_{z+\Delta z} + q_{out} V_b = \text{mass rate storage}$$

C. L. - 1

$$[p_v]_x - [p_v]_{x+\Delta x} + [p_v]_y - [p_v]_{y+\Delta y} + [p_v]_z - [p_v]_{z+\Delta z} + [q_{in} V_b - q_{out} V_b] = \text{mass rate storage}$$

Reau porosity is defined as $\phi = V_p / V_b \rightarrow V_p = \phi V_b$ where

$V_b = A \Delta z$ But the mass stored in the pore volume

$$= P V_p = P \phi V_b$$

$$\text{Therefore, Mass rate storage} = \frac{\partial}{\partial t} (P \phi V_b)$$

$$[p_v]_x - [p_v]_{x+\Delta x} + [p_v]_y - [p_v]_{y+\Delta y} + [p_v]_z - [p_v]_{z+\Delta z} + \Delta z + [q_{in} V_b - q_{out} V_b] = \frac{\partial}{\partial t} (P \phi V_b)$$

Applying Taylor's series.

$$f(x + \Delta x) = f(x) + \Delta x \frac{\partial f}{\partial x}$$

$$f(x) - f(x + \Delta x) = -\Delta x \frac{\partial f}{\partial x}$$

$$\text{This implies that } [p_v]_x - [p_v]_{x+\Delta x} = -\Delta x \frac{\partial p_v}{\partial x}$$

$$\therefore -\Delta x \frac{\partial (p_v)}{\partial x} - \Delta y \frac{\partial (p_v)}{\partial y} - \Delta z \frac{\partial (p_v)}{\partial z} = \frac{\partial}{\partial t} (P \phi V_b)$$

$$+ [q_{out} V_b - q_{in} V_b]$$

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Note: $q_{net} = q_{in} - q_{out}$.

$$q = Av$$

$$-\Delta x \frac{\partial (P u_x A_x)}{\partial x} - \Delta y \frac{\partial (P u_y A_y)}{\partial y} - \Delta z \frac{\partial (P u_z A_z)}{\partial z} =$$

$$\frac{\partial}{\partial t} (P \psi V_b) + q_{net} V_b.$$

Applying transport equation from Darcy's

$$u_x = -B_c \frac{k}{\mu} \left(\frac{\partial p}{\partial x} + \gamma_1 \frac{\partial z}{\partial x} \right)$$

Substitute

Hence:

$$\Delta x \frac{\partial}{\partial x} \left[B_c \frac{k P A_x}{\mu} \left(\frac{\partial p}{\partial x} - \gamma_1 \frac{\partial z}{\partial x} \right) \right] + \Delta y \frac{\partial}{\partial y} \left[B_c \frac{k P A_y}{\mu} \left(\frac{\partial p}{\partial y} - \gamma_1 \frac{\partial z}{\partial y} \right) \right] + \Delta z \frac{\partial}{\partial z} \left[B_c \frac{k P A_z}{\mu} \left(\frac{\partial p}{\partial z} - \gamma_1 \frac{\partial z}{\partial z} \right) \right] = \frac{V_b}{\alpha_c} \frac{\partial}{\partial t} (P \psi) +$$

$$q_{net} V_b.$$

Dividing through by $\text{Vol}(B)$ we have $\frac{V_b}{\alpha_c}$

$$\Delta x \frac{\partial}{\partial x} \left[B_c \frac{k P A_x}{\mu B_1} \left(\frac{\partial p}{\partial x} - \gamma_1 \frac{\partial z}{\partial x} \right) \right] + \Delta y \frac{\partial}{\partial y} \left[B_c \frac{k P A_y}{\mu B_1} \left(\frac{\partial p}{\partial y} - \gamma_1 \frac{\partial z}{\partial y} \right) \right] + \Delta z \frac{\partial}{\partial z} \left[B_c \frac{k P A_z}{\mu B_1} \left(\frac{\partial p}{\partial z} - \gamma_1 \frac{\partial z}{\partial z} \right) \right] = \frac{\rho V_b}{\alpha_c} \frac{\partial}{\partial t} \left(\frac{\psi}{B_1} \right) + q_{net} V_b$$

Assuming the same density then.

$$\Delta x \frac{\partial}{\partial x} \left[B_c \frac{k A_x}{\mu B_1} \left(\frac{\partial p}{\partial x} - \gamma_1 \frac{\partial z}{\partial x} \right) \right] + \Delta y \frac{\partial}{\partial y} \left[B_c \frac{k A_y}{\mu B_1} \left(\frac{\partial p}{\partial y} - \gamma_1 \frac{\partial z}{\partial y} \right) \right] + \Delta z \frac{\partial}{\partial z} \left[B_c \frac{k A_z}{\mu B_1} \left(\frac{\partial p}{\partial z} - \gamma_1 \frac{\partial z}{\partial z} \right) \right]$$

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Considering the right hand side of the equation about applying product rule.

$$\frac{\partial}{\partial t} (P\phi) = P \frac{\partial \phi}{\partial t} + \phi \frac{\partial P}{\partial t} \quad (\text{differentiate both respect to } P)$$

$$\frac{\partial}{\partial t} (P\phi) = P \frac{\partial \phi}{\partial P} \frac{\partial P}{\partial t} + \phi \frac{\partial P}{\partial P} \frac{\partial P}{\partial t}$$

Factor out

$$\frac{\partial}{\partial t} (P\phi) = P\phi \left(\frac{1}{\phi} \frac{\partial \phi}{\partial P} + \frac{1}{P} \frac{\partial P}{\partial P} \right) \frac{\partial P}{\partial t}$$

Recall

$$C_r = \frac{1}{\phi} \frac{d\phi}{dP} \quad (\text{Pore compressibility}) \text{ of the rock formation.}$$

$$C_f = \frac{1}{P} \frac{\partial P}{\partial P} \quad (\text{fluid compressibility.})$$

$$\frac{\partial}{\partial t} (P\phi) = (P\phi)(C_r + C_f) \frac{dP}{dt}$$

$$C_t = C_r + C_f$$

$$\frac{\partial}{\partial t} (P\phi) = P\phi C_t \frac{\partial P}{\partial t}$$

$$\Delta x \frac{\partial}{\partial x} \left[B_c \frac{k P \Delta x}{\mu} \left(\frac{\partial P}{\partial x} - y_1 \frac{\partial z}{\partial x} \right) \right] + D_y \frac{\partial}{\partial y}$$

$$\int \left[B_c \frac{k P \Delta y}{\mu} \left(\frac{\partial P}{\partial y} - y_1 \frac{\partial z}{\partial y} \right) + D_z \frac{\partial}{\partial z} \left(B_c \frac{k P \Delta z}{\mu} \left(\frac{\partial P}{\partial z} - y_2 \frac{\partial z}{\partial z} \right) \right) \right]$$

$$= \frac{V_b}{\alpha_c} P\phi C_t \frac{\partial P}{\partial t} + q_{net} V_b$$

PROVEN!!!

7

Paper 2 of 3 — PEE 523 PQ

Level 500 • Second Semester • PastQuestion

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DEPARTMENT OF PETROLEUM ENGINEERING, FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
PRODUCTION ENHANCEMENT & OPTIMIZATION PNG 523 CLASS TEST 9/9/19
(WRITE ON THE QUESTION PAPER) AUGUST 2017

TIME: 1 HOUR

NAME: AKHIBI DAVID

MAT NO 1501/2013

- 1) A well 10,000 ft deep was completed several years ago without gaslift mandrels. It has quit production due to the decline in the reservoir pressure and increase in watercut. Management has approved that the well can be put on 'poor boy' gaslift. The current reservoir pressure is 3000 psig. The SG of the reservoir fluid is 0.729. Graphically design the 'poor boy' gaslift system. Assume that liftgas gradient is 0.01 psi/ft and its casing injection pressure is 1200 psig.
- What is the liquid level?
 - At what depth should the gas be injected?
 - If a drawdown of 400 psi is attained and the aerated column gradient is 0.15 psi/ft, what is the flowing wellhead pressure?

Depth = 10,000 ft
 $P_{res} = 3000 \text{ psig}$
 Liftgas gradient = 0.01 psi/ft
 Specific gravity of reservoir fluid = 0.729
 Casing Injection Pressure = 1200 psig
 Aerated column gradient = 0.15 psi/ft
 Drawdown Pressure = 400 psi

Solution
 i) for Liquid Level $P = 0.0052 \text{ CA}$

$$\Delta = \frac{P_r}{0.433 \times S.G.} = \frac{3000}{0.433 \times 0.729}$$

$$\Delta = 9503.98 \text{ ft}$$
 Liquid Level = TVD - 9503.98 ft
 $= 10,000 - 9503.98 \text{ ft} = 496 \text{ ft}$ 20

- ii) If the Drawdown is 400 psi: $3000 - 400 = 2600 \text{ psi}$
 Assuming Depth of 4500 ft and 5500 ft to find FTHP
 $0.15 \times 0.433 \times 4500 = 292.275 \text{ psi}$
 $0.15 \times 0.433 \times 5500 = 373.58 \text{ psi}$
 $2600 - (292 + 373.6) = 1934.4 \text{ psi}$
 2) An IPO gaslift valve having the properties given below is to be installed in a mandrel of a well at depth 4000 ft. Bellows area is 1.0 sq in. Port area is 0.2 sq in. The valve opening pressure in the laboratory is 1000 psig. The casing pressure at 4000 ft is 1350 psig while the tubing pressure is 500 psig. Recommend a dome pressure for this valve. What is the valve opening pressure?

$A_b = 1.0 \text{ sq in.}$
 $A_p = 0.2 \text{ sq in.}$
 Casing Pressure = 4000 ft @ 1350 psig
 Tubing Pressure = 500 psig
 $P_{voo} = 1000 \text{ psig}$

$$R = \frac{A_p}{A_b} = \frac{0.2}{1.0} = 0.2$$

$$TEF = \frac{R}{1-R} = \frac{0.2}{(1-0.2)} = \frac{0.2}{0.8}$$

$$TEF = 0.25$$

$$P_{vo} = P_{voo} - P_f \times TEF$$

$$P_d = R P_f + P_{voo} (1-R)$$

$$P_{vo} = 1000 - (500 \times 0.25)$$

$$P_{vo} = 1000 - 125$$

$$P_{vo} = 875 \text{ psig}$$
 20

$$P_d = R P_f + P_{voo} (1+R) \quad \text{At Lab } P_f = 0$$

$$P_d = 0.2(500) + 1000(1-0.2)$$

$$P_d = 100 + 1000(0.8)$$

$$P_d = 920 \text{ psig} + 800 \text{ psig} = 1720 \text{ psig}$$
 20

Dome Pressure is 800 psig, while the Valve opening Pressure = 875 psig

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 PRODUCTION ENHANCEMENT & OPTIMIZATION / PNG 523 CLASS TEST
 (WRITE ON THE QUESTION PAPER) AUGUST 2017

TIME: 1 HOUR

NAME: HASALI HARDEE A.

MAT NO: 12451271

- 1) A well 10,000 ft deep was completed several years ago without gas lift in adms. It has quit production due to the decline in the reservoir pressure and increase in watercut. Management has approved that the well can be put on 'phoratory' system. The current reservoir pressure is 3000 psig. The SG of the reservoir fluid is 0.8. Graphically design the 'phoratory' system. Assume that lift gas gradient is 0.01 psi/ft and its casing injection pressure is 1000 psig.
- What is the liquid level?
 - At what depth should the gas be injected?
 - If a drawdown of 400 psi is attained and the aerated column is 15.025 ft, what is the flowing wellhead pressure?

~~Liquid level = $\frac{10000}{0.8} = 12500$~~

~~When $P_o = 0.425 \times 0.8 = 0.34$~~

~~Liquid level = $\frac{10000}{0.34} = 29411$~~

Check back for a above

- 2) A well was severely damaged during drilling and completion. A build-up survey carried out on it confirmed a skin factor of 15. A carefully planned and executed stimulation treatment reduced the skin to 1. The reservoir, well and fluid data are shown below. Use the Darcy equation to estimate the pre and post stimulation flow rates for the same flowing bottom hole pressure.

Reservoir Pressure	= 3800 psig	Flowing Bottom Hole Pressure	= 3000 psig
Reservoir Thickness	= 300 ft	Reservoir Permeability	= 250 md
Reservoir Fluid Viscosity	= 30 cp	External Drainage Radius	= 750 ft
Well Bore Radius	= 5.125 inches	Formation Volume Factor	= 1.2 res bbl/stb
API Gravity of Crude	= 30°		

Pre-Stimulation rate

$$Q = \frac{2\pi kh \Delta P}{\mu L \left[\ln \left(\frac{r_e}{r_w} \right) + \frac{1}{2} + S \right]}$$

$$= \frac{2 \times 3.14 \times 42 \times 300 \times (3800 - 3000) \times 0.25}{30 \times 1.2 \left[\ln \left(\frac{750}{5.125} \right) + 0.5 + 15 \right]}$$

$$= \frac{377040}{790.95}$$

$$= 476.69 \text{ stb/day}$$

Post stimulation flow rate = 476.69 stb/day

Post Stimulation rate

$$Q_{psb} = \frac{2\pi kh \Delta P}{\mu L \left[\ln \left(\frac{r_e}{r_w} \right) + \frac{1}{2} + S \right]}$$

$$= \frac{2 \times 3.14 \times 42 \times 300 \times (3800 - 3000)}{30 \times 1.2 \left[\ln \left(\frac{750}{5.125} \right) + 0.5 + 1 \right]}$$

$$= \frac{377040}{286.95}$$

$$= 1313.95 \text{ stb/day}$$

Post stimulation flow rate = 1313.95 stb/day

O.A.P's Perry
100
100

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EFFURUN
PRODUCTION ENHANCEMENT & OPTIMIZATION PNG 523 CLASS TEST
(WRITE ON THE QUESTION PAPER) SEPT 2018

TIME: 1 HOUR

NAME: ADENIYI PAUL OLORUNSHOLA MAT NO 1470/2013

Study the attached gaslift design for a well having the following data below;
Well Depth = 10,000 ft
Oil NGR gradient = 0.350 psi/ft
Dome is Nitrogen charge. The correction factor can be assumed as $1 + 0.000028D$, where D is the GLV depth.
Bellows area is 0.31 sq in.
Tubing Size = 3 1/5" OD, 2.99 in ID
GL Valves are 1" PPO with 3/16" port
Port area is 0.029 sq in.

Calculate the following:

- The pressures the valve domes should be set in the laboratory at standard conditions.
- The valve opening and closing pressures.
- The well flowing bottom hole pressure.

Please show your answers in a table.

Well Depth (ft)	Correction Factor	F _{oi} (psi)	P _{oi} (psi)	TRO (psi)	P _{vo} (psi)	P _{vc} (psi)
1700	1.0476	420	875	442.29	373.04	420.00
2600	1.0729	500	875	514.17	461.20	500.00
3350	1.0938	580	875	584.98	549.14	580.00
4030	1.1128	640	875	634.46	615.24	640.00
4780	1.1316	700	875	682.43	679.36	700.00
5350	1.1498	760	875	729.20	745.55	760.00
5950	1.1666	820	875	775.44	809.88	820.00
6250	1.175	880	875			

$$0 = P_f \left(\frac{A_b}{A_b - A_p} \right) - C_t$$

$$= \frac{A_b}{A_b - A_p} = \frac{0.31}{0.31 - 0.029} = 1.1032$$

$$0 = TRO \times C_t = P_c \left(\frac{A_p}{A_b - A_p} \right)$$

$$\frac{A_p}{A_b - A_p} = \frac{0.029}{0.31 - 0.029} = 0.1032$$

$$P_{vc} = TRO \times C_t / \left[\frac{A_b}{A_b - A_p} \right]$$

Where TRO - Pressure of the Valve domes should be set in the Lab
P_{vo} - Opening Pressures

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COLLEGE OF TECHNOLOGY

2018/2019 Second Semester Examination
PNG 523: Production Enhancement & Optimization

Answer All Questions

Time: 3 Hours

Please write your Matriculation Number Boldly on your Script

- 1a) Define the following Petroleum Production Optimization terms:
- I) Low Production
 - II) Off Production
 - III) Completion Interval Potential
 - IV) Reservoir Well Potential
 - V) Conduit Potential

- 1b) A well's test data and current production level (CPL) are the same but management insists that the well should be optimized. Use a flowchart to illustrate how this well should be optimized.

- 2a) The USA is using the technology of fracturing of tight shale reservoirs has now become a net exporter of petroleum. Describe in detail the application of this technology.

- 2b) A well was damaged during drilling and completion. A build-up survey carried out on it confirmed a skin factor of 16. Acid Stimulation will reduce the skin completely. The reservoir, well and fluid data are shown below. Use the Darcy equation to estimate the pre and post stimulation flow rates for the same flowing bottom hole pressure.

Reservoir Pressure	= 3600 psig	Flowing Bottom Hole Pressure	= 3000 psig
Reservoir Thickness	= 300 ft	Reservoir Permeability	= 400md
Reservoir Fluid Viscosity	= 40 cp	External Drainage Radius	= 750 ft
Well Bore Radius	= 5.125 inches	Formation Volume Factor	= 1.1 resbbl/stb
API Gravity of Crude	= 35°		

- 3a) What is Artificial Lift? How is it different from secondary recovery processes?

- 3b) An Indigenous Company has acquired a depleted well which was abandoned by an International Oil Company and wants to put it on intermittent gaslift. The well data are as follows:

Well Depth (Mid Perforations)	= 10,000 ft
Foot Valve Depth	= 9,000 ft
Gaslift Valve Depth	= 8,000 ft
Slug Height	= 2,000 ft
Oil Gravity	= 0.7
Gradient of pure water	= 0.433 psi/ft
Lift gas gradient	= 0.01 psi/ft

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1. Well Intervention is the re-entering of well for remedial or repair activities

Reasons for Well Intervention

(1) Equipment Failure

* Corrosion
* Wear of moving parts

(2) Formation related Issues

* Impairment due to workover operation
* Gaslift or water deposit
* Inflow due to drilling fluid

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Surface Injection Pressure = 2500 psig

Graphically determine the following:

- The Reservoir Pressure
- The flowing tubing head pressure when the gas arrives at the Injection Gaslift Valve
- The flowing tubing head pressure when the slug arrives at the surface (i.e. the top of the slug is at the surface)

4a) What is well Intervention? Discuss briefly the two major reasons for well Intervention.

4b) Management has decided to put a well on continuous gaslift and you have been hired to carry the design given the following specifications:

Valve Type	PPO (Production Pressure Operated)
Min FWHP	100 psig
Kick-off Pressure	1100 psig
Kill Brine Gradient	0.465 psi/ft
Liftgas Gradient	0.01 psi/ft
Oil with NGLR Gradient	0.350 psi/ft
Well Depth	10,000 ft
Tubing Size	3 1/2"

Below is the Equilibrium Curve

Depth, ft	Pressure, psig
0	100
1000	140
2000	220
3000	340
4000	480
5000	670
6000	870
7000	1120
8000	1380
8200	1420

- Determine the mandrel depths graphically.
- What is the flowing bottom hole pressure?

5) If the bellows area and the port area of the valve are 0.35 sq in and 0.030 sq in respectively in your design in Question 4b, recommend a setting pressure (TRO) of the valve in each mandrel. The temperature correction factor may be assumed as $1 + 0.000042D$ where D is the valve depth.

$$P_{br} = \frac{P_{fAb}}{A_b - A_p}$$

$$P_{br} = \frac{P_{fAb}}{A_b - A_p} \cdot \frac{1}{C_t}$$

$$TRO = \frac{P_{br}}{C_t} = \frac{P_{fAb}}{C_t(A_b - A_p)}$$

$$TRO = \frac{P_{fAb}}{C_t(A_b - A_p)}$$

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2017/2018 Second Semester Examination
PNG 523: Production Enhancement & Optimization

Answer All Questions

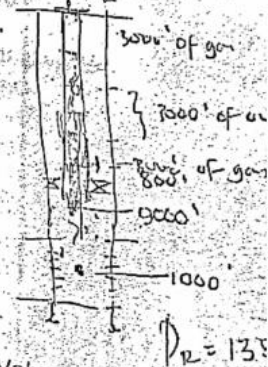
Time: 3 Hours

Please write your Matriculation Number Boldly on your Script

- 1a) Define Completion Interval Potential. Explain how re-perforation can be used to optimize this Potential.
- 1b) The well test data and current production level (CPL) have been found to be very close. Use a flowchart to show how production from this well can be optimized.

- 2a) Define Intermittent and Continuous Gaslift.
- ✓ 2b) A well was abandoned by a Multi-National Oil Company because it is depleted. However, an Indigenous Company acquired it and put it on Intermittent gaslift. The well data are as follows:

Well Depth (Mid Perforations)	10,000 ft
Foot Valve Depth	9,000 ft
Gaslift Valve Depth	8,000 ft
Slug Height	3,000 ft
Oil Gravity	0.8
Gradient of pure water	0.433 psi/ft
Liftgas gradient	0.01 psi/ft
Surface Injection Pressure	2500 psig



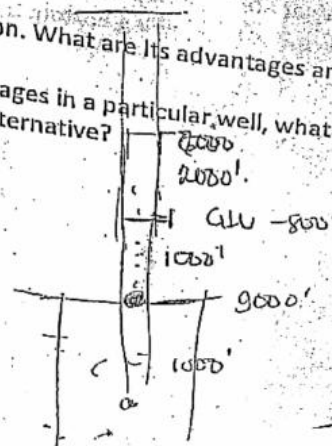
Graphically determine the following:

- The Reservoir Pressure
- The flowing tubing head pressure when the gas arrives at the Gaslift Valve
- The flowing tubing head pressure when the slug arrives at the surface (i.e. the top of the slug is at the surface)

- 3a) What is Beam Pumping? Briefly describe the operation. What are its advantages and disadvantages?
- 3b) If you cannot use Beam Pumping due to its disadvantages in a particular well, what alternative can you use in its place? Why have you chosen this alternative?

$$P_{wf} @ 8000' = 2500 + 0.01 \times 8000$$

$$= 2580$$



Scanned with CamScanner

- 4a) What is acid washing in well stimulation?
 4b) What is chemical stimulation without acid?
 4c) A well was damaged during drilling and completion. A build-up survey carried out on it confirmed a skin factor of 10. A carefully planned stimulation program was poorly executed and the skin was increased to 15. The reservoir, well and fluid data are shown below. Use the Darcy equation to estimate the pre and post stimulation flow rates for the same flowing bottom hole pressure.

Reservoir Pressure	= 3500 psig	Flowing Bottom Hole Pressure	= 250 psig
Reservoir Thickness	= 200 ft	Reservoir Permeability	= 30 md
Reservoir Fluid Viscosity	= 35 cp	External Drainage Radius	= 750 ft
Well Bore Radius	= 5.125 inches	Formation Volume Factor	= 1.2 res bbl/stb
API Gravity of Crude	= 30°		

- 5) Management has decided to put a well on continuous gaslift and you have been hired to carry out the design given the following specifications:

Valve Type	IPO (Injection Pressure Operated)
Min FWHP	100 psig
Kick-off Pressure	1100 psig
Kill Brine Gradient	0.465 psi/ft
Liftgas Gradient	0.01 psi/ft
Oil with NGLR Gradient	0.360 psi/ft
Well Depth	10,000 ft
Tubing Size	3 1/2"

Below is the Equilibrium Curve

Depth, ft	Pressure, psig
0	200
1000	240
3000	340
4000	450
5000	600
6000	800
7000	1020
8000	1280
9000	1590

- i) Determine the number of mandrels including the orifice.
 ii) What are their depths?
 iii) What is the flowing bottom hole pressure?

Paper 3 of 3 — PEE 523 PQ

Level 500 • Second Semester • PastQuestion

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'ARTMENT OF PEROLEUM ENGINEERING, Corin IVERSITY OF PETROLEUM RESOURCES, EFFUR'

oerion ENHANCEMENT & OPTIMIZATION PNG 523 CLASS TEST 9/9/19

PRopU (WRITE ON THE

QUESTION PAPER) AUGUST 2017

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NAME AKH IB Bawid matno - 15

} |Awell 10,000 ft deep was completed st

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c everal years 880 without

the reservoir pressure and increase in,

TIME: 1 HOUR

Baslift mandrels. It has quit Production due to the ecline in

e watercut, Management has approved that the well can be put on Poor boy' gas Hin

The current reservolr pressure #3000 psig. the sq of the reservoir fluid is 0.729. Graphically design the 'poor boy' gas

m. Assume that liftgas gradient 1S 0.01 psistt andi ing injecti sure is 1200 psig.

[syste én What is the liquia level its casing injection pressu

i ii) At what depth should the gas be Injected?

i iii) fa drawdown of 400 psi is attained and the aerated column gradient is 0.15 psi/ft, what is the flowing

| wellhead Pressure? . .

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 stalied'in'a mandrel of alwell at depth 4000 ft." SADE 6p
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 8 Pressure is 500 psig. Recommend a dome Pressure fo
 ftis 1350 psi r this valve, What is
 6 Pressure?
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fi PRODUCTION NPANCE MENT Primi, TION . 23 CLASS TEST

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df sme:1 Hour "NAME: ae ARE bag

[TME:IH . Mew, fay DLORY AL stHoxY MAT No "tho /eo/a 7

'Study the attached gastirt design for 4 well having the following data below; Bee

Well Depth = 10,000 f ; . Tubing Size = 3 2/5" OD, 2.99 in 15 ..

Oil NGR gradient * 9.350 psi/ft "GL Valves ara 1" PPO with P/16"port*

Dome is Nitrogen charge The CORrecti factor ϕ , be assy das1+ 9.00002ap, Where pj.

| theGIV depth. . oo Port area ig g 29sq in,

i Bellows area is 0.34 Sqin, : -f

Calculate the following: :

il) The va}

"domes Should be set}

Ve Opening ang Closing Pressures

: iti) The welt flowing bottoin hole

Please show your answers jn a table.

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] i) The Pressures the valve

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FEDERAL UNIVERSITY of PETROLEUM RESOURCES, EFFURUN

Y COLLEGE OF TECHNOLOGY af

2018/2019 Secong Semester Examination =

PNG 523. FOduction En ancement & Optimization % ϕ

Answer All Questions Time: 3 Hours

Please Write your atriculation Number Boldly on your Script

la) Define the following Petroleum Oduction Optimization terms:

) low Production

1) Off Toductj ~~,

MI) Completion Intervay Potential . ie

lv) Reservoir w, 'oten Bie

v) Conduit Potential a

1b) A well's test data and Current Production level (CPL) are the same but management insists that the well should be optimized, Use a flowchart to illustrate how this Well should be optimized,

2a) The well is using the technology of fracture stimulation, so now become an net exporter of petroleum, it is a;

2b) A well was damaged during a shut-in survey carried out on it and confirmed a skin factor of 16, which is completely, The reservoir is a "well" and fluid data are shown below, estimate the stimulation flow rates for the same

Reservoir Pressure

Reservoir Thickness

Reservoir Fluid Viscosity

Initial Pressure = 3000 Psig

Initial Flow Rate = 400 m³/d

External Drainage Radius = 750 ft

Well Bore Radius = 5.125 inches Formation Volume Factor = 1.15 bbl/stb

API Gravity of Crude = 35° API :

3a) What is Artificial Lift? How is it different from Secondary recovery Processes? ~

3b) - An indigenous community has acquired a depleted well which was abandoned by an international oil company and wants to put it on intermittent gas lift. The well data are as follows: a)

Well Depth (True Vertical Depth) = 10,000 ft

Foot Valve Depth = 9,000 ft. : :

Gaslift Valve Depth = 8,000 ft —<

Slug Height = 2,000 ft

Oil Gravity = 0.7

Gradient of pure Water = 0.433 psi/ft

; Lift gas gradient of 0.01 psi/ft : ing 8 WOR!

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. £. +4. 2500 psig

oe Surface Injection Pressure na ES :

. Graphically determine the following:

. Graphically determine the following:

i) The Reservoir Pressure

lacti lift Valve

' Injection Gas

ii) The flowing tubing head pressure when the gas arrives at the Inj

. top of the

a urface (i.e. the

ii) The flowing tubing head pressure when the slug arrives at the s

slug is at the surface) .

. j ion.

for ré or well Interventi

What is well Intervention? Discuss briefly the two major réasons f

: 4 een hired to carry

Ja) Management has decided to put a well on continuous gaslift and you have b :

the design Biven the fi

ollowing specifications: . ; 7

Valve Type PPO (Production Pressure Operated):

Min FWHP 7

100 psig - . *

Kick-off Pressure 1100 psig ee ge ee a y

Kill Brine Gradient 0.465 psi/tt ts esse ;

Liftgas Gradient

0.01 psi/ft

Oil with NGLR Gradient 0.350 psi/ft "GI ne ;

Well Depth 10,000 ft = 8 8Y :

Tubing Size ' "31/2" : :

Below is the Equilibritim Curve ° an aaa

r Depth, ft i

ve See

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" ili) What is the flowia

5) If the bellows area'ang the pot ares'

design in Question 4b, recommend a temperature Corre

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of. the valve are 0.38 sq in,

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ere'D is the valve depth':

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COLLEGE 9

_ Answer all

Please w;

Questions © - Time: 3 Hours... a

rite your. Matriculation Number Boldly on Your Script -

» , Define Completion Interval Potential, Explain how re.

12) "Potential, . — . . ee

The well test data and current production level (cpp

tr flowchart to show how Production from this Well ca

da)

Vb)' :

it and PUTtoF

d termine the following:

a. The Reservoir Presstire

3a).

3b)

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Chosen this

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'Sanna th CamSeonne

4a) - What Is acid washing iii well ocean ;

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What is chemical stimulation wit hou a ; wilde

4b) . "A well was damaged during drilling and completion. Ab Le vias pootly executed and
 4c) confirmed a skin factor of 10. A carefully planned stimulation prog Mt a ow' Use the Damrey
 the skin was increased to 15. The reservolr, well and fludl data are sh » flowy bottom hole
 equation to estimate the. pre and post stimulation flow rates for the same,
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pressure. : '

Reservoir Pressure. =3500 psig Flowing Bottom Hole Pressure = aa0 psig

* Reservoir Thickness . . = 200ft Reservoir Permeability = 30% nq}

* Reservoir Fluid Viscosity 35.cp.-/ «External Drainage Radiyis © 5 750.

Well Bore Radius "=: 5.425 inches Formation Volume Factor 21.21 /6th

~ API Gravity of Crude. =30" a0 boos. i '

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5). Management has decided to put a well on contin

voi. the design given the-following specifications:

: Valve Type © PO (Inj

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Kick-off Pressure'

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Below isthe Equawrium Curve: .

's Mepth, ft : Pressure, psig

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